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Realizing the potential of tidal currents and the efficiency of turbine farms in a channel

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Abstract

Tidal turbines in strong flows have the potential to produce significant power. However, not all of this potential can be realized when gaps between turbines are required to allow navigation along a channel. A review of recent works is used to estimate the scale of farm required to realize a significant fraction of a channel's potential. These works provide the first physically coherent approach to estimating the maximum power output from a given number of turbines in a channel. The fraction of the potential realizable from a number of turbines, a farm's fluid dynamic efficiency, is constrained by how much of the channel's cross-section the turbines are permitted to occupy and an environmentally acceptable flow speed reduction. Farm efficiency increases as optimally tuned turbines are added to its cross-section, while output per turbine increases in tidal straits and decreases in shallow channels. Adding rows of optimally tuned turbines also increases farm efficiency, but with a diminishing return on additional rows. The diminishing return and flow reduction are strongly influenced by how much of the cross-section can be occupied and the dynamical balance of the undisturbed channel. Estimates for two example channels show that realizing much of the MW potential of shallow channels may well be possible with existing turbines. However unless high blockage ratios are possible, it will be more difficult to realize the proportionately larger potential of tidal straits until larger turbines with a lower optimum operating velocity are developed.

Highlights

► The economics of adding new turbines to a row spanning the width of a channel are better than the economics of adding new rows of turbines to a farm. ► Adding rows to a tidal turbine farm reduces the power produced per turbine, even though total power production may increase. ► Realizing the maximum power available in a small tidal channel may be possible with existing turbines, if they almost fill the channel's width. ► Realizing the Giga Watt potential of larger tidal straits will be difficult unless much of the strait's width can be filled with turbines.

Introduction

Underwater turbines in strong tidal currents can contribute to the need for renewable energy sources. To make a significant contribution many turbines must be grouped into large tidal farms to exploit the energy of high flows through narrow tidal channels. This leads to a critical question, how does power production scale as a turbine farm expands from a single turbine into a large energy farm? While adding turbines to a farm increases its generation capacity, power extraction also enhances drag on the flow along the channel. This enhanced drag slows flows along the channel, reducing power production by all turbines. Flow reduction becomes more significant as a farm grows and ultimately limits a channel's potential to produce power ([1], hereafter GC05). So it is the combined effects of flow reduction and installed generation capacity which determines how much power an expanding farm produces.

Part of addressing the question of how power production scales is estimating the maximum power a particular channel can generate, i.e. the channel's "potential" (see review [2]). Most recent estimates of potential have used the approach developed by GC05 (e.g. [1]). To realize that potential requires the turbines to occupy the entire cross-section of the channel, have perfect electro-mechanical efficiency, that there are no losses to mixing behind the turbines and that there is no drag on any structure supporting the turbines. Realistically none of these are achievable, however GC05's potential provides a useful upper bound for production.

The need for navigation along a channel will often restrict turbines to only part of the channel's cross-section, Fig. 1. Thus just as important as estimating a channel's potential is to determine how much of a channel's potential can be realized from a given number of turbines when some flow can bypass the turbines through gaps left to permit navigation, Fig. 2. In addition, flow reduction may also have environmental impacts, such as reduced tidal exchange and sediment transport (e.g. [4]). Thus another constraint on turbine numbers is likely to be an acceptable flow speed reduction. This requires answering the question, how does power production scale with farm size within constraints on how much of the cross-section can be filled with turbines and how much flows within the channel can be reduced?

An older method for estimating a channel's potential was to use the Kinetic Energy Flux through the undisturbed channel and an arbitrary loading factor of 10–15% to estimate power output from the channel [2]. However the KE flux through a channel without turbines is unrelated to a channel's potential as it does not account for the flow reduction due to power extraction [5]. Also, the KE flux does not provides a means to estimate the power available from a given number of turbines which may occupy only part of the cross-section, nor a means to estimate the flow reduction. Both these aspects are critical in assessing the tidal current resource at a particular site, the farm's impacts and the economics of developing it. Building on GC05 and GC07 the recent [6], [7] (hereafter V10 and V11) works provide the first physically coherent approach to maximizing the power output from a given number of turbines. The upper bound for the power available for electricity production given by these models can be used to answer the question about how power production scales within constraints, as well as how best to arrange and configure the turbines.

In this work a review of GC05 [8], (hereafter GC07), V10 and V11, is used to discuss the question and indicate the scale of farm required to realize a significant fraction of a channel's potential. How much of a channel's potential can be realized, i.e. farm efficiency, is strongly linked to turbine efficiency. Not only the turbine's electro-mechanical energy conversion efficiency, but also the turbine's fluid dynamic efficiency. This work focuses on fluid dynamic efficiency, thus addressing the "power available" for electricity production from a farm with a given number of turbines. Turbines extract energy from the flow passing through the area spanned by their blades. To maximize turbine efficiency all turbines must have the strength of the flow passing through them adjusted or tuned, typically achieved by varying blade pitch. To maximize farm output the flow speed through the turbines must be tuned for a particular channel and fraction of the channel's cross-section taken up by the turbines, as well as tuned in relation to each other (V10 and V11). This makes maximizing the fluid dynamic efficiency of a farm complicated.

The four fundamental works, GC05, GC07, V10 and V11, are necessarily mathematical and complex. This work aims to firstly review the essential results from these works to make them accessible to a wider audience and then use the results to discuss the number of turbines required to realize a significant fraction of the potential of two example channels. This work begins by reviewing the physics of tidal channel potential and turbine efficiency contained in the works by outlining four key concepts in Section 2. Understanding the physics of farm efficiency hinges on understanding how the farm's gross drag coefficient affects the flow along the channel. Section 3 looks at the efficiency of a single row of turbines. Section 4 examines the effects of adding more rows. Section 5 looks in detail at how many turbines are required to achieve a significant fraction of the example channels' potentials when constrained by cross-sectional occupancy and flow reduction.

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Section snippets

Background physics

The physics underlying the latter sections are reviewed in the next four sub-sections, each of which presents an essential concept. The underlying aim of this section is to make the connection between the farm's drag coefficient and the power which is available for electricity production clear. The farm's drag coefficient is the link between the number of turbines in the farm and the power available for electricity production. The essential idea is that the farm's drag coefficient increases as ...

Farm fluid dynamic efficiency

A measure of a farm's efficiency is the fraction of GC05's potential which is available for power production, V10, i.e. $\mathbf{FE} = \frac{P_{farm}}{P_{GC05}}$

FE is the farm's fluid dynamic power efficiency and is maximised at the optimal tuning, α^{opt}, V11. This is the headline efficiency, which is the upper bound for the fraction of a channel's estimated potential that can be turned into electricity from a given number of turbines.

Fig. 4a) shows FE for the two example channels as optimally tuned turbines are ...

Efficiency of multi-row farms

Adding capacity to a farm by adding rows of optimally tuned turbines increases farm efficiency as demonstrated in Fig. 5a). For the tidal strait 15 rows makes only 40% of it potential available, so as would be expected, the associated flow reduction is modest. In contrast for the shallow channel 3 rows make 70% of its potential available, with a significant associated flow reduction.

Fig. 5 illustrates a second extreme way to realize most of a channel's potential, having a large number of rows, ...

Realizing the potential: a trade off

The decision on turbine numbers is an economic trade off between the income from the power made available by adding turbines to a farm against their cost and any environmental impacts. Once the maximum permissible cross-sectional occupancy has been achieved then the trade off is in the face of a diminishing return on additional turbines, Fig. 5a). How this trade off plays out depends strongly on the maximum fraction of the channel's cross-section which can be occupied in order to maintain ...

Discussion

The principle aim of this paper was to show how many turbines are required to realize a given fraction of a channel's potential in order to improve understanding of how power production scales with farm size. The models in V10 and V11, which build on those of GC05 and GC07, provide the first physically coherent way of connecting the number of turbines in a farm with the maximum power available for electricity production. The models do not require the use of arbitrary loading or safety factors ...

Conclusions

Critical questions in developing a farm are what fraction of a channel's potential is available for power production from a given number of turbines as a farm scales up from a single turbine into a large farm. The V10+V11 works provide the first physically coherent way to connect the number of turbines in a farm to both the maximum power available for electricity production and to the degree of flow reduction, which is a consequence of power extraction. Though simple they illuminate several ...

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...The need for individualized tuning arises because turbines in channels interact with each other, even if they are so far apart that their wakes do not impinge on each other. This interaction results from the enhanced drag due to power extraction by a turbine, reducing the free-stream flow for every turbine in a channel [V10, 10,11]. Section 3.2. Fig. 5. Optimally tuned turbines in large arrays perform very differently to tuned isolated turbines: For example, they may have optimal power and thrust coefficients much higher, or lower, than at the Betz limit....
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